Predicting Ski Jumps using State-Space Model

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Outline

- Introduction
- Modeling Framework and Dataset
  - State-space models and Least squares method
  - Data processing
- Methodology
  - Different approaches and error function
- Main Results
  - Animation app
- Discussion and Conclusion

Motivation

- Planning of ski jump enlargement and anticipating risky conditions
- Goal: Predict ski jumps based on external factors

Figure: Flying hill in Planica

- State-space models (SSM)
- Least squares method (LS)

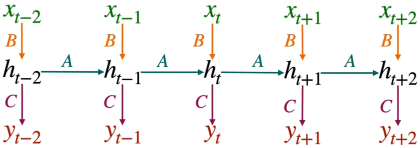


Figure: Schema of SSM

Customized caption

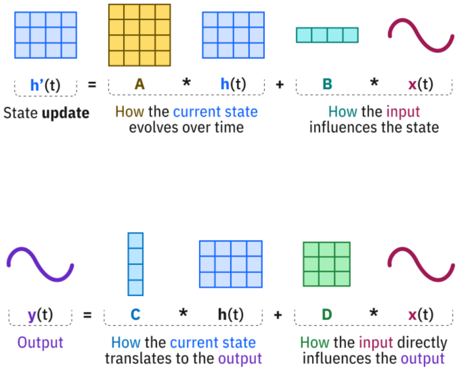


Figure: My first figure in beamer.

Data Description

- Data structure
- Angles
- coordinate system on a main ski jumping hill in Planica

Angles

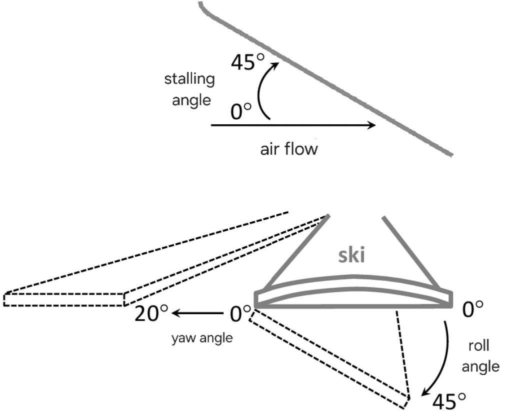


Figure: Representation of different measured angles.

Ski jumping hill

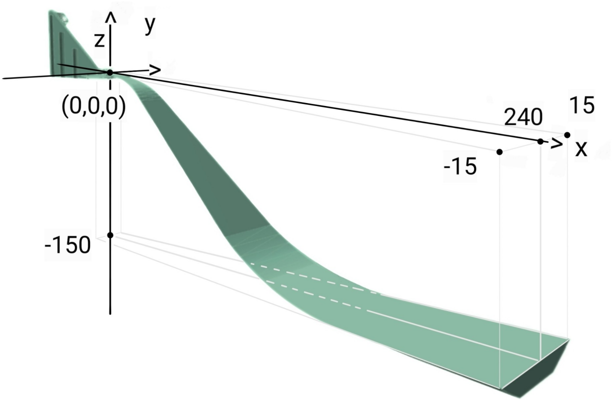


Figure: Ski jumping hill with a coordinate system placed on top.

Different approaches and error function

- pure SSM
- SSM + PINN (Physics Informed Neural Networks)

Main Findings

Key Result

Your important result goes here.

Flight simulation example

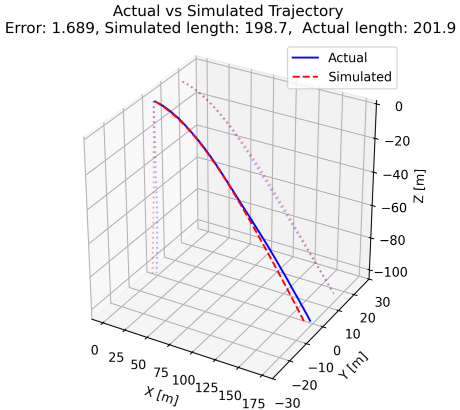


Figure: Graph with a simulated and actual flight trajectory.

Application

- Description of the application
- Observing measured and predicted jumps

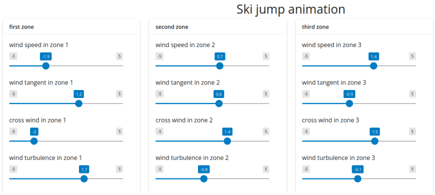
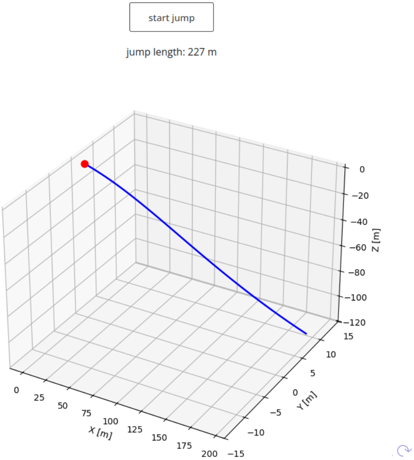


Figure: Adjusted sliders in the app



Interpretation

- Limitations
- Future work and potential improvements
- Short recap

Thank you!